



FRACTIONS - COMPARING & ORDERING UNLIKE FRACTIONS

's Learn Together!

What Are Fractions?

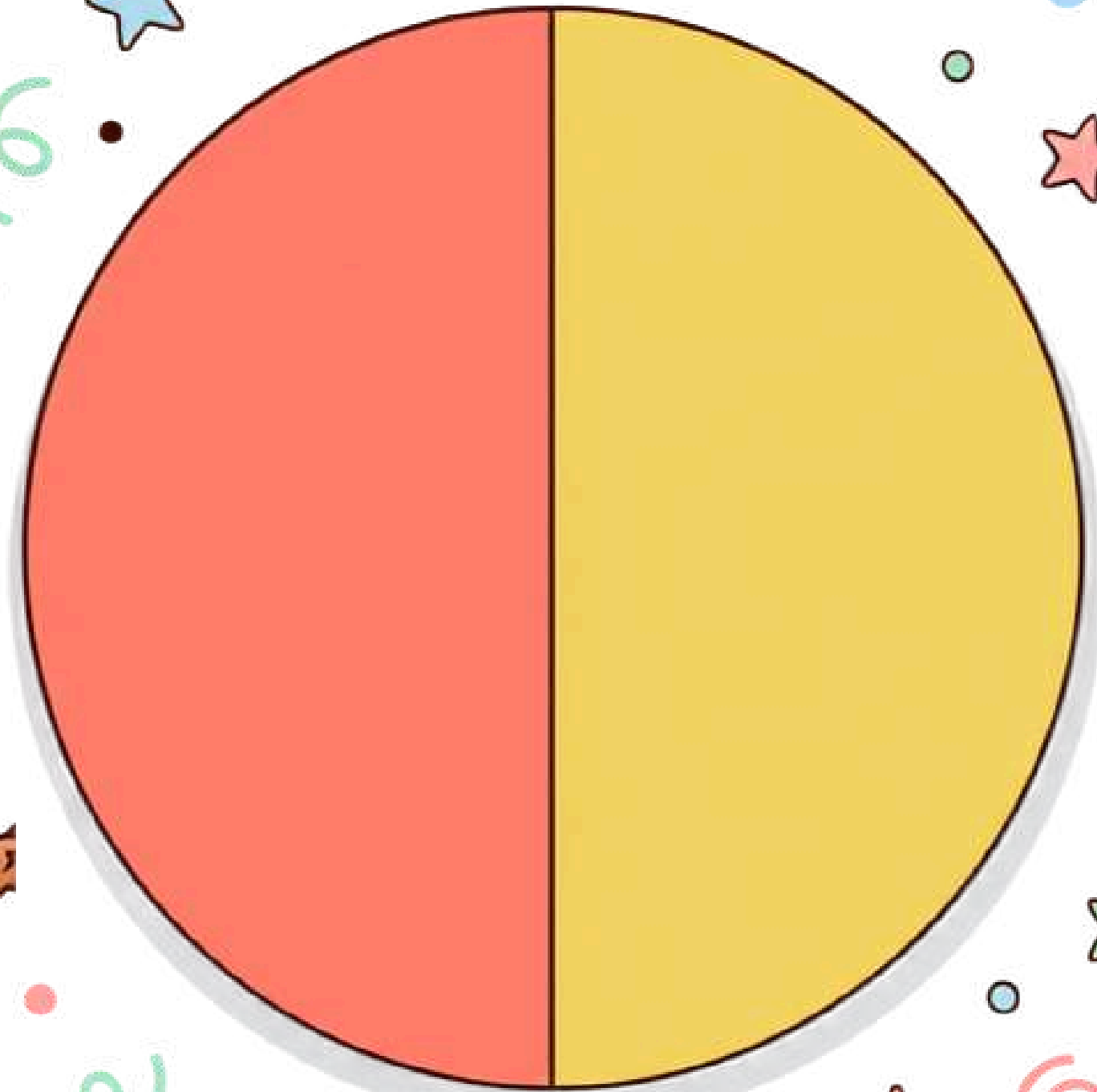
A fraction shows a part of a whole.

Example: $\frac{1}{2}$ means one part out of two equal parts.

The top number (numerator) is how many parts we have.

The bottom number (denominator) is how many equal parts the whole is split into.

Fractions help us share things fairly and measure accurately!



WHY DO WE COMPARE FRACTIONS?

We compare fractions to know which is bigger or smaller.

Examples:

- Who got more pizza?
- Which glass has more juice?
- Who ran a longer distance?

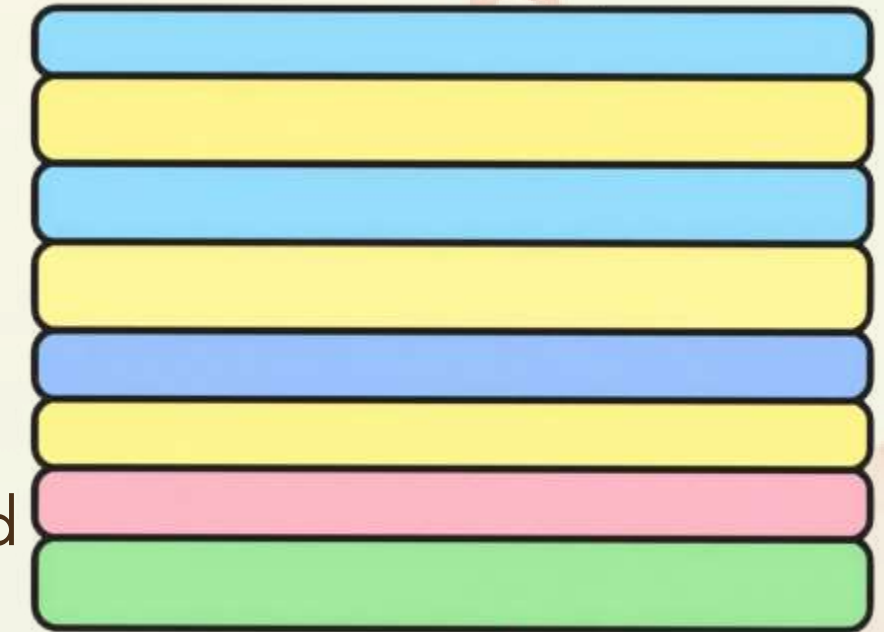


WHAT ARE UNLIKE FRACTIONS?

Unlike fractions have different denominators (bottom numbers).

Examples: $\frac{1}{2}$, $\frac{1}{3}$, $\frac{2}{5}$

They show parts of a whole, but the whole is divided into different numbers of pieces!



VISUAL MODELS HELP US UNDERSTAND

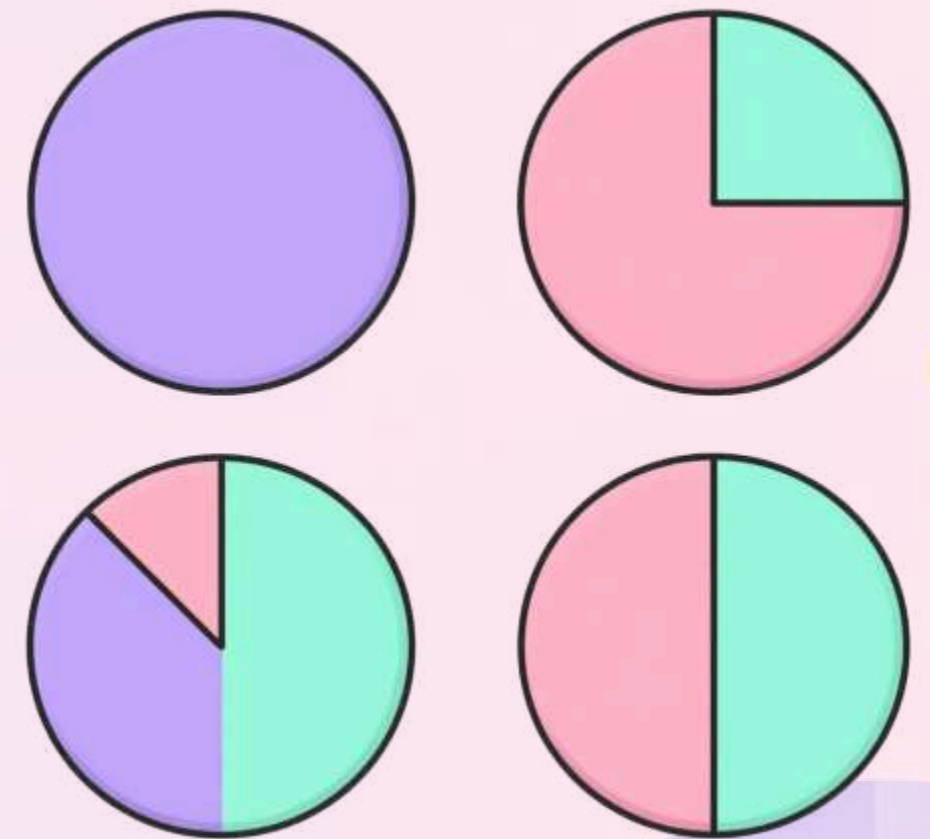
Fraction bars help us SEE which fraction is bigger!

Halves: divided into 2 equal parts

Thirds: divided into 3 equal parts

Fifths: divided into 5 equal parts

The more parts, the smaller each piece gets!



HOW CAN WE COMPARE UNLIKE FRACTIONS?



STEP 1: FIND COMMON DENOMINATOR

Look at the bottom numbers of your fractions and find a number both can divide into equally.

STEP 2: MAKE EQUIVALENT FRACTIONS

Change each fraction so they have the same denominator. Multiply top and bottom by the same number.

STEP 3: COMPARE NUMERATORS

Now compare the top numbers! The bigger numerator means the bigger fraction.



EXAMPLE 1: COMPARING $\frac{1}{2}$ AND $\frac{1}{3}$



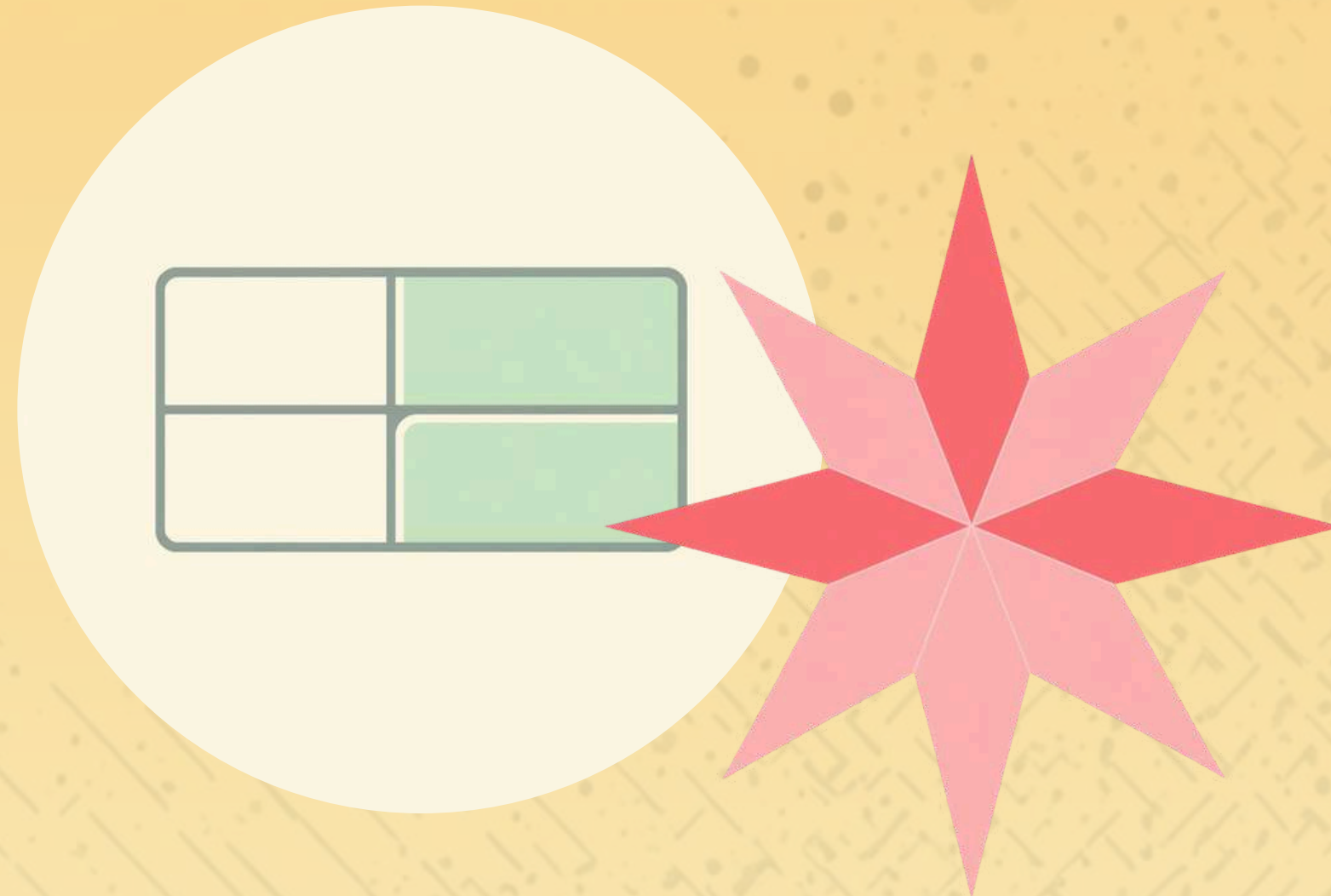
Convert to same denominator:

$$\frac{1}{2} = \frac{3}{6} \text{ and } \frac{1}{3} = \frac{2}{6}$$

Compare numerators: $3 > 2$

So, $\frac{1}{2} > \frac{1}{3}$

The fraction bar shows $\frac{1}{2}$ is bigger!



ORDERING 3 FRACTIONS: $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{3}$

Find a common denominator (12) to compare these fractions. Convert each fraction, then order them from smallest to largest!

HOW
LESS THAN

$$\frac{1}{4} =$$



$$\frac{1}{3} = \frac{1}{3}$$

$$\frac{1}{2} = \frac{1}{2}$$



AND 6 SYSS.



PRACTICE TIME!
WHICH IS BIGGER?
 $\frac{2}{5}$ OR $\frac{3}{10}$?
HINT: FIND A COMMON
DENOMINATOR!

VISUALIZING FRACTION SIZES

SEE HOW FRACTIONS LOOK IN REAL
LIFE!

Looking at real objects like cake slices helps us understand and compare fractions easily. When we can see the sizes, it becomes much simpler to tell which fraction is bigger or smaller!





TIPS FOR COMPARING FRACTIONS



FIND COMMON DENOMINATOR

Always find a common denominator first before comparing any fractions. This makes the fractions easier to compare!

USE FRACTION BARS

Draw fraction bars or pictures to see which fraction is bigger. Visual models help you understand better!

COMPARE NUMERATORS

After converting, the bigger numerator means a bigger fraction. Always check your work by drawing!

PRACTICE QUESTION 2

$$\frac{3}{8}$$

$$\frac{1}{2}$$

$$\frac{5}{5}$$

Order smallest to largest:

$\frac{3}{8}$, $\frac{1}{2}$, $\frac{5}{8}$

Can you solve it?

Hint: Find a common denominator

Convert all fractions

Compare the numerators

COMMON MISTAKES TO AVOID

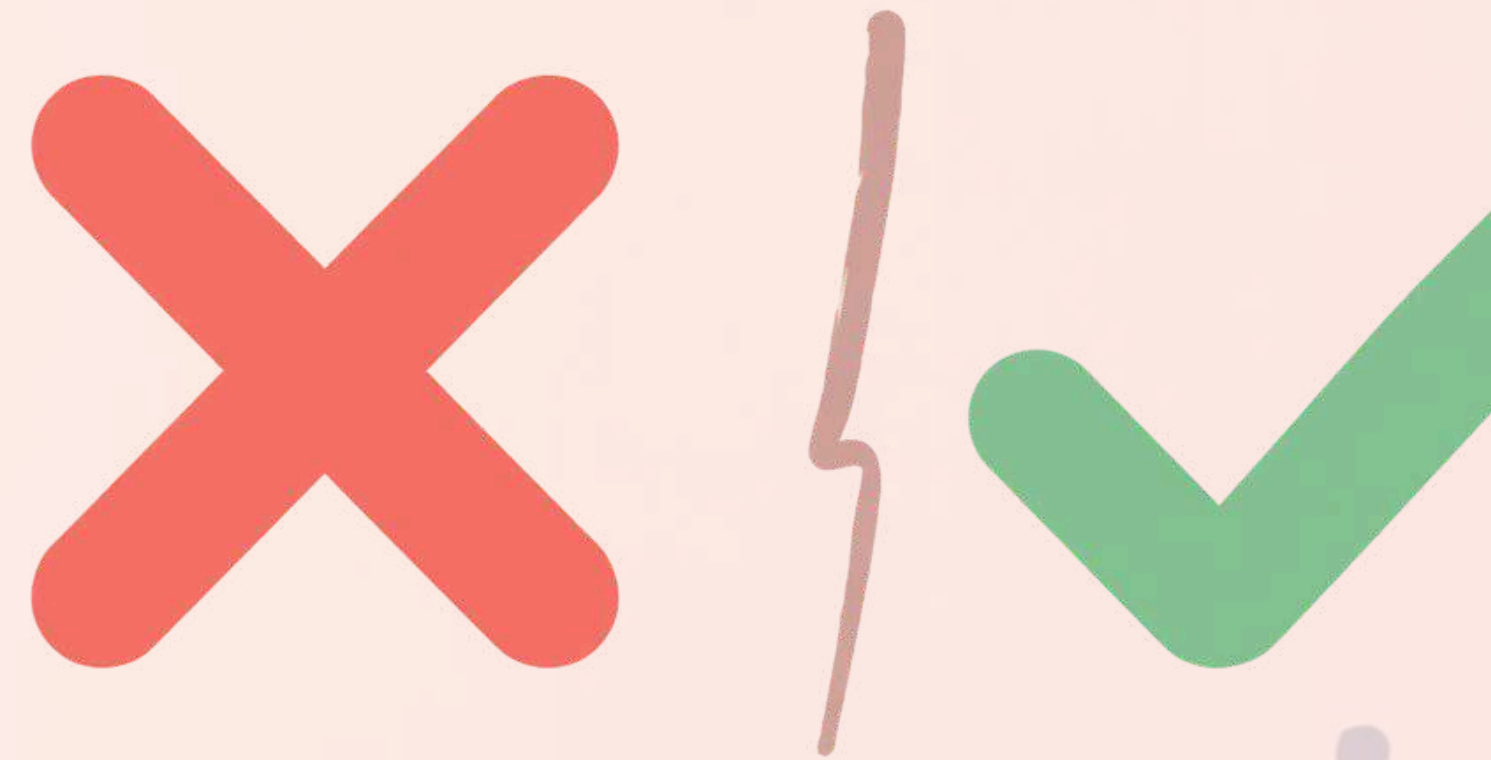
Mistake 1: Comparing numerators without finding a common denominator first.

Mistake 2: Thinking larger denominators mean bigger fractions ($1/4$ is NOT bigger than $1/3$ just because $4 > 3$).

THE RIGHT WAY

Wrong: $1/4 > 1/3$ because $4 > 3$

Right: Convert first! $1/4 = 3/12$ and $1/3 = 4/12$, so $1/3 > 1/4$





RECAP: STEPS TO COMPARE &

ORDER UNLIKE FRACTIONS

FIND COMMON DENOMINATOR

Look for the smallest number that all denominators divide into evenly.

CONVERT FRACTIONS

Change each fraction to an equivalent fraction with the common denominator.

COMPARE NUMERATORS

Look at the top numbers — the bigger numerator means the bigger fraction!



SUMMARY & ENCOURAGEMENT

You've learned so much today! Remember, comparing fractions is easy when you follow the steps: find a common denominator, convert, and compare. Keep practicing and you'll become a fractions expert!

"Math is fun when we take it one step at a time!"





THANK YOU!

Any questions? Math is fun when we take it one step at a time!

